ment mainly because their Wi-Fi is occupied for other applications. Once again, keeping this in mind, medical devices are being designed with multiple wireless platform connectivity modes," he adds.

Essential design tips

Medical device designs need to meet several regulations and pass safety compliance tests. Often, designers face the challenge of rework after putting up a device that misses some safety regulations.

Nitesh explains, "All components (including regulators and capacitors) should be certified or approved by regulatory agencies like ISO and IEC. This may add to the cost, but it is a necessity. If your design uses radio-components, these should also have a regulatory certificate."

Appropriate hardware component selection is crucial. While sensors vary from device to device depending on the intended application, microcontroller remains the brain of computation. Choosing the right microcontroller is imperative to designing a device appropriate for its purpose.

"In life-critical devices, processors with backup cores should be preferred. In case of a failure in the main core, the alternate core will keep the device running. For example,

Fig. 2: Connectivity modules like Bluetooth and Wi-Fi are increasingly being included in medical device design

Pre-built modules for quicker design

As noted by many manufacturers, designers can speed up their work by integrating pre-developed modules. These are quick-to-integrate and pre-approved medically. However, in the absence of indigenous development, these components are usually imported.

Nitesh says, "Instead of designing your own hardware completely, you can use certified multiple modules for your product. These modules are mainly imported. This is a challenge in terms of cost control. Yet, it adds an advantage in terms of time-to-market since the modules are readily available from distributors like Digikey or Mouser. The reason why such components are not being made in India is the lack of demand. There are very few manufacturers of electronic medical devices in India. So sufficient demand volume is not generated."

ARM Cortex R-series are becoming the microcontrollers of choice. That's because these have an in-built safety mechanism that gets activated in case of any failure," says Nitesh.

Nitesh adds that companies like Texas Instruments are building MCUs exclusively for medical devices. Many medical devices require low-amplitude data signals, say, 24-bit, and that is the kind of design these controllers follow.

Another important design aspect is safety. Electrical insulation should also be very thorough.

Jangir says, "You should know the amount of insulation required for your hardware. For instance, for a specific test, the device should stay stable at 3kV line voltage. Moreover, if you are designing a high-speed device, you should know how to properly ground the overall circuit. This will keep any kind of radiated emission (RE) as low as possible. Additionally, the hardware needs to be robust enough for any situation."

The devices should be made electromagnetic-interference (EMI)-proof as much as possible. Electromagnetic compliance (EMC) can be ensured through electrical safety, mechanical safety and EMI safety.

Like any other electronic equipment, medical devices are prone to overheating. To make the device heat-resistant, first ensure proper grounding. Next, choose an appropriate heat-sink. Additionally, ensure sufficient ventilation in the design schematic. In the least likely scenario, you will need to introduce a fan into the device.

Test and measurement tools can ensure precision of medical data. For a customised medical device design, get your device tested and certified. Software for medical devices are mostly

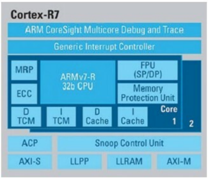

Fig. 3: An example: The ARM Cortex R7 architecture (Image source: rtcmagazine.com)

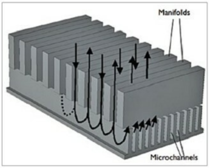

Fig. 4: Representative image of how heat-sinks work (Source: comsol.com)

embedded. There are predefined regulations that guide developers to build software optimal for data safety and proper functioning.

To sum up

Compliance with defined regulations and selection of appropriate hardware components are crucial for optimal circuit design. With the inclusion of more powerful hardware, better processing units with safety protocols and wireless connectivity, the ever-evolving medical equipment keep getting safer and more reliable. It would be interesting to see how these further incorporate even newer and more intelligent technologies. EFY